

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 5 : Session – 2 : SLO – 2 | Graph Representation — Adjacency List

■ I. Fill in the Blanks

- In an adjacency list, each vertex in a graph is associated with a LINKED LIST (list of its neighbours).
- The adjacency list is more efficient in terms of space for SPARSE graphs.
- The time complexity for checking if there is an edge between two vertices in an adjacency list is $O(\text{degree})$ / $O(V)$ worst case.
- In an undirected graph, if there is an edge between vertex A and vertex B, then both A and B will appear in each other's ADJACENCY LIST.
- The space complexity of an adjacency list representation is $O(V + E)$.

■ II. True or False

- Adjacency list is preferred for representing dense graphs — FALSE (matrix is preferred for dense graphs).
- In a directed graph, each edge is stored in both vertices' adjacency lists — FALSE (stored only at the source).
- Adjacency list can store weights along with the edges in a weighted graph — TRUE.
- Traversing all neighbours of a vertex in an adjacency list is faster than in an adjacency matrix for sparse graphs — TRUE.

■ III. Short Answers

- Adjacency list: an array of linked lists; each vertex stores a list of all vertices adjacent to it.
- Advantages: (1) memory efficient for sparse graphs, $O(V + E)$ space; (2) fast iteration over a vertex's neighbours.

■ IV. Adjacency List — Undirected Graph (edges A-B, A-C, B-D, C-D, B-E, D-E)

- A -> B, C
- B -> A, D, E
- C -> A, D
- D -> B, C, E
- E -> B, D

■ V. Adjacency List — Directed Graph (A->A, A->B, A->C, B->E, D->B, D->C, D->E)

- A -> A, B, C
- B -> E
- C -> (none)
- D -> B, C, E
- E -> (none)

■ VI. Solve and Write — 4-Vertex Cycle (edges 1-2, 2-4, 4-3, 3-1)

- Adjacency list:
- 1 -> 2, 3
- 2 -> 1, 4
- 3 -> 1, 4
- 4 -> 2, 3
- Vertices adjacent to vertex 3: vertices 1 and 4.
- Number of edges: 4.